

SEMINAR REPORT:

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CLASS: AIDS-B

SUB: PTIS

DESIGN OF EXPERIMENTS

Outlook Statement on Applications:

The various designs of Analysis of Variance (ANOVA) — including Completely Randomized Design (CRD), Randomized Block Design (RBD), and Latin Square Design (LSD) — are used for planning and analyzing experiments across various fields.

- In agriculture, they help evaluate the influence of factors like fertilizer and irrigation methods while minimizing environmental impact.
- In medical and clinical research, they are used to assess treatment effectiveness and accounting for confounding factors.
- Industrial and manufacturing sectors employ these designs to optimize production processes, improve quality control, and test machine or material performance.
- Social sciences and education also benefit, enabling the evaluation of teaching methods, learning environments, or psychological treatments with controlled variables.

One-way Classification – Completely Randomized Design (CRD)

Applications:

- Product testing: Comparing customer satisfaction with different product features.
 - Agriculture: Studying the effect of one type of fertilizer on crop yield.
- Lab experiments: Testing different chemical treatments on a solution to observe reaction rates.
 - Industrial trials: Evaluating the effect of one machine setting on production speed.
 - Textile testing: Analyzing different fabric treatments for durability.
- Education research: Comparing student performance using different teaching methods.

Two-way Classification – Randomized Block Design (RBD)

Applications:

- Agriculture: Testing multiple fertilizers (treatments) on different soil types (blocks).
- Medical trials: Comparing treatment effects across different patient age groups or hospitals.
- Food industry: Testing packaging material and storage temperature on shelf life.
- Construction: Studying the effect of cement type on strength across different sites.
- Market research: Testing advertising methods in different regions (blocks).
- Manufacturing: Studying the impact of raw material type and operator skill on product quality.

Three-way Classification – Latin Square Design (LSD)

Applications:

- Agronomy: Studying three factors like crop variety, fertilizer, and irrigation method while controlling for row and column variation (e.g., land layout).
- Biological sciences: Experiments where subjects are tested under three conditions with limited repetitions.
- Animal science: Measuring feed efficiency for different diets across breeds and stalls.
- Packaging industry: Studying packaging type, label design, and storage location.
- Psychology research: Evaluating treatments across time slots, patients, and therapists.
- Factory trials: Testing machine settings, operators, and shifts using minimal resources.

